Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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1. Introduction

This report presents the analysis of a Si–SiO_2 double quantum dot device simulated with QTCAD. Starting from the electrostatic model used in the previous laboratory, the numerical workflow was extended by solving the single-particle Schrödinger equation in the quantum-dot region after the nonlinear Poisson problem had converged.

The report first describes the modifications introduced in the Python script, including the computation of the first five quantum states with the prescribed solver tolerance. The generated output files are then analysed to verify the carrier depletion in the active region, the formation of two confinement minima, the localization of the eigenstates, and the coupling configuration of the two dots.

2. Changes in python script

In order to solve the Schrodinger equation for the device analyzed in this laboratory, we modified the given code whose initial aim was uniquely to calculate the potential inside the device throught the non-linear Poisson equation.

```
# Schrodinger configuration
    num_states = 5
    params_schrod = SchrodingerSolverParams()
    params_schrod.num_states = num_states
    params_schrod.tol = 1e-9
```

Listing 2.1: Initialization of Schrodinger parameters

First (2.1), we create the object `SchrodingerSolverParams()`, a class that will contain all the parameters that we want to set. Then, we defined the number of energy levels to consider to be equal to 5: this will tell to the program to consider only the first 5 solutions of the Schrodinger equation, storing the eigenfunctions with their correspective eigenvalues. Another paramater that is set is the threshold in the convergence process during the calculation: this value, which in this case is set to be equale to $1e-9$, indicates the degree of convergence we want to achieve in defining a solution valid: if the difference between two subsequential iteration for the same value is less than this threshold, then the value will be considered converged and so valid.
Then, after defining the parameters for the solver, the following commands are added:

```
    d.set_V_from_phi()
    submesh = SubMesh(d.mesh, dot_region)
    subd = SubDevice(d,submesh)

    schrod_solver = SchrodingerSolver(subd, solver_params=params_schrod)
    schrod_solver.solve()
```

Listing 2.2: Physical definition of the domain

- `d.set_V_from_phi()`: This command updates the internal state of the `Device` object `d` by populating a new attribute, the potential energy array `V`. Physically, it performs a node-by-node mapping of the electrostatic potential ϕ into the potential energy V using the relation $V(\mathbf{r}) = -q \cdot \phi(\mathbf{r})$. Without this step, the Schrödinger solver would lack the necessary confinement potential to find bound states.

- `SubMesh` and `SubDevice`: These commands initialize the phyisical domain where the Schrodinger equation will be calcualted. By defining a `submesh` limited to the `dot_region`, the simulation focuses computational resources only on the volume where the quantum wavefunctions are expected to be non-zero. In fact, `dot_region` in an array defined previously in the code containing all the names of Phyisical Groups of our interest, which are:

```
dot_region=["region_pre_quantum_B_mid","region_quantum_B_mid",
            "region_post_quantum_B_mid"]
```

Listing 2.3: Regions containing the quantum dots

  The `SubDevice` object `subd` inherits all physical properties (such as temperature and materials) from the main device but operates on this significantly smaller mesh, drastically reducing memory usage and computation time.

- `SchrodingerSolver` and `.solve()`: These commands initialize and execute the numerical solver for the time-independent Schrödinger equation. The solver transforms the physical Hamiltonian operator into a large, sparse matrix using the Finite Element Method (FEM), subsequently solving the resulting eigenvalue problem $H\psi = E\psi$.

After the calculation, the object `subd` will contain the first 5 solutions of the Schrodinger equation, that will subsequentially extracted and used to generate the output file `DQD_SiMOS_q.vtu` and `DQD_SiMOS_energies.txt`. The first is used in Paraview in order to visualize the shape and value of the wave functions, while in the second text file are reported the eigenvalues (energy values in eV) of the relative eigenfunction.

3. Analysis of the output files

The generated outputs were inspected at both the electrostatic and quantum levels. The `n` file is used to verify the carrier distribution and the depletion of the active region, while the `q` file is used to analyse the confined single-particle states.

3.1 Analysis of the n file

To evaluate the spatial distribution of carriers before the quantum population regime, a two-dimensional slice of the electron charge density n is extracted at the coordinate $Z = 45$ nm. This specific height is crucial as it accurately cuts through the active silicon channel where the double quantum dot is designed to form.

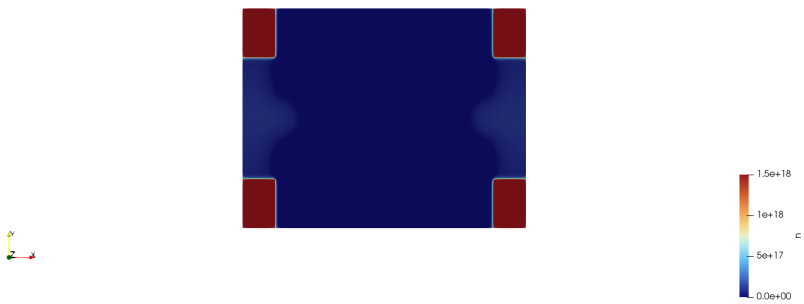


Figure 3.1: Slice at Z = 45 of the file n.

As shown in the 2D density map in Fig. 3.1, the central region of the channel is completely depleted of electrons. This behavior is directly governed by the electrostatic action of the lateral barrier gates. These gates have a direct influence on the behavior of the reservoirs and the two dots: in fact, the shape of the Drain and the Source, represented by the light blue regions at the extremities of the photo, exhibits a small protrusion towards the center, which is tightly constrained from both below and above along the Y-axis. This is the direct effect of the Y-gates, which are strictly required to achieve electrostatic confinement for the quantum dots along the Y-direction as well. Consequently, the carriers can only communicate with the central dots by tunneling through the narrow potential barriers.A distinctive feature of the density map is the presence of four highly populated regions located at the corners of the domain, highlighted in dark red due to their maximum electron density values. The inclusion of these highly doped n-type zones is crucial for the operation of the device at cryogenic temperatures (10mK). At such extreme temperatures, standard silicon undergoes the phenomenon of **carrier freeze-out**, losing its conductivity and turning into an insulator. This would prevent the formation of proper ohmic contacts with the external metal lines due to the creation of insurmountable Schottky junctions.To bypass this limitation, these four corner regions are heavily doped with n-type impurities, shifting the material into a degenerate state where it behaves like a metal, maintaining a high and stable concentration of free electrons even at millikelvin temperatures. These zones act as intermediate electron reservoirs connected to the external contacts. When the gate voltages are applied, these highly conductive corners ensure that carriers can be readily supplied to the classical Source and Drain extensions (the light blue horizontal channels) and, subsequently, tunnel into the central quantum dots. To quantitatively validate these 2D observations, a 1D Plot Over Line was performed across the symmetry axis at $Y = 136.5$ nm.

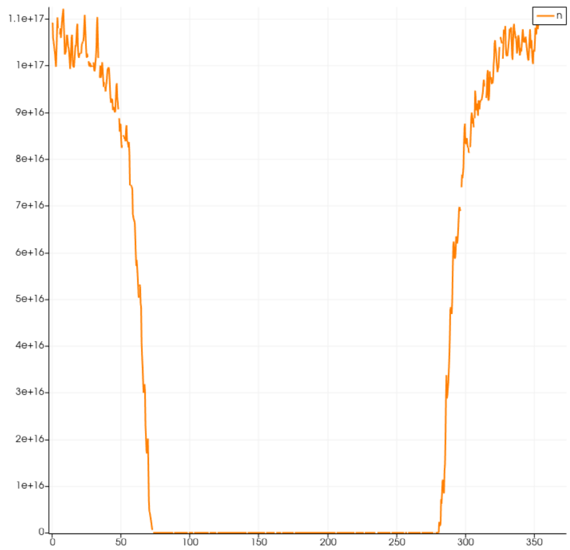


Figure 3.2: Plot overline at Y = 136.5 of the file n.

The 1D profile provides a clear confirmation of the assumptions made before: the electron density reaches its maximum values at the opposite extremes of the domain, marking the horizontal Source and Drain reservoirs cut by the line. Specifically, we can notice that this maximum concentration is on the order of 10^{17}cm^{-3}, which represents a high value, yet remains lower than the peak concentration of 10^{18}cm^{-3} previously observed in the degenerate corner regions. Conversely, the density drops sharply and remains strictly zero throughout the entire central core. This confirms that the electrostatic landscape successfully isolates the quantum dot region from classical drift-diffusion currents, establishing the ideal zero-background environment required for single-particle Schrödinger-Poisson simulations.

3.2 Analysis of the `q` file

The analysis of the `DQD_SiMOS_q.vtu` output was carried out in ParaView by cutting the device at the height of the quantum layer ($z\!=\!45$nm) and by extracting a *Plot Over Line* along the axis connecting the two dots ($y\!=\!136.5$nm). The plotted quantities are the confinement potential and the probability densities $|\psi_i|^2$ of the first five single-particle eigenstates.

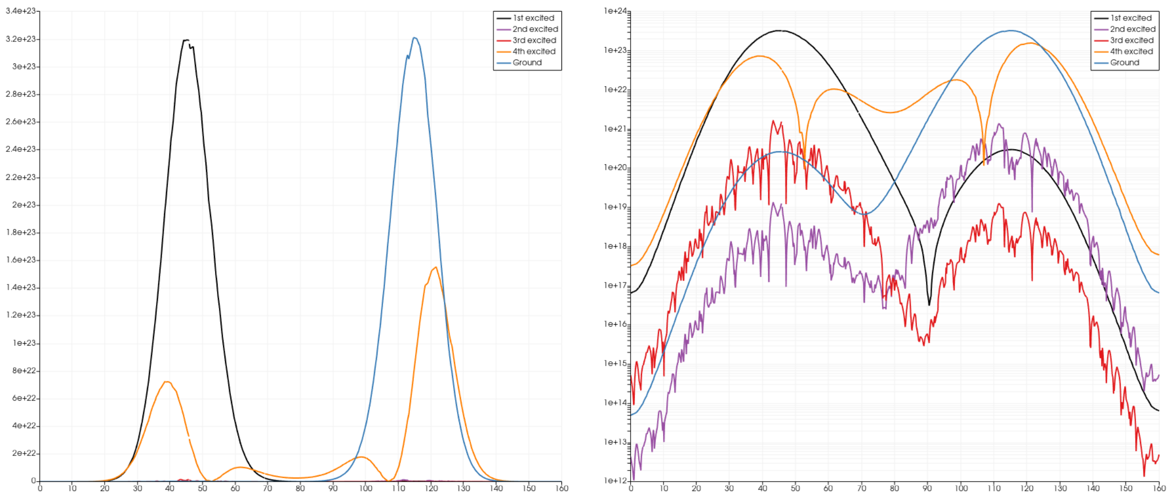
State	Energy [eV]	Localization observed in the quantum slice
Ground	0.01115999	Ground orbital of the right dot.
1st excited	0.01120989	Ground orbital of the left dot.
2nd excited	0.01396172	First excited orbital of the right dot.
3rd excited	0.01400675	First excited orbital of the left dot.
4th excited	0.01438458	Higher orbital with finite density in the barrier region.

Table 3.1: First five eigenenergies obtained from the Schrödinger solution.

For an ideal symmetric double dot, the two states belonging to the same orbital shell should be degenerate. In the simulation they instead appear as two close but resolved pairs. The ground-state pair is split by $4.99\times 10^{-5}\text{eV}\!=\!49.9\mu\text{eV}$, while the first excited pair is split by $4.50\times10^{-5}\text{eV}\!=\!45.0\mu\text{eV}$. These splittings are small compared with the energy separation between the two shells, $E_2-E_0\simeq 2.80$meV. We therefore interpret the spectrum as two almost degenerate dot orbitals, staggered by a small left-right asymmetry. This staggering is already present at the level of the initial geometry/mesh, so it is reported as a numerical/geometrical splitting of the nominally degenerate states rather than as an additional physical conclusion from the post-processing.

Figure 3.3: Confinement potential on the quantum slice. The two minima define the two dots, while the central saddle gives the tunnel barrier.

The confinement map in Fig. 3.3 shows two electrostatic minima separated by a finite central barrier. Sampling the same line used for the wavefunction plots gives $V_L \simeq 5.03$meV and $V_R \simeq 4.84$meV for the two dot minima, and an inter-dot saddle of $V_b \simeq 15.83$meV. The fourth excited level, $E_4 = 14.38$meV, is therefore only about 1.45meV below the barrier top. This explains why this state has a visible probability density inside the barrier region, while the lower levels remain more strongly confined inside the individual wells.

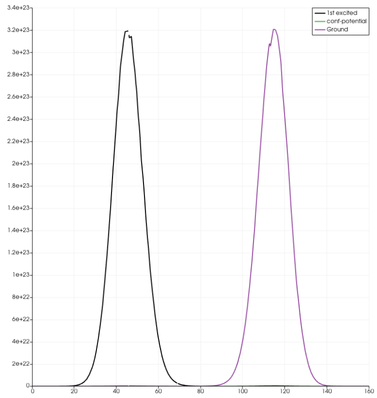


(a) Linear scale.

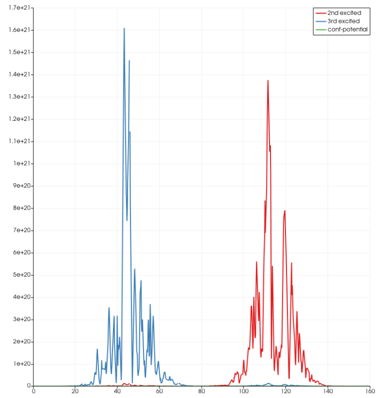
(b) Logarithmic scale.

Figure 3.4: Probability densities along the axis connecting the two dots. The logarithmic representation is more informative because it resolves the weak tails and the small excited-state densities that are hidden on the linear scale.

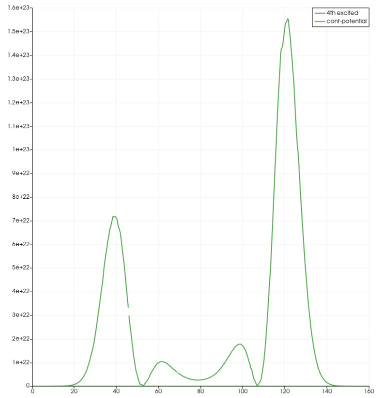
The line profiles in Fig. 3.4 show the hierarchy of the states. On the linear scale the two lowest orbitals dominate because their peak densities are approximately two orders of magnitude larger than the second and third excited states along the selected line. The logarithmic plot keeps the same spatial information but makes the lower-amplitude components visible, in particular the tails across the inter-dot region.



(a) Ground-state pair.



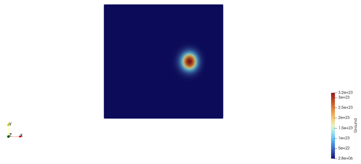
(b) First excited pair.



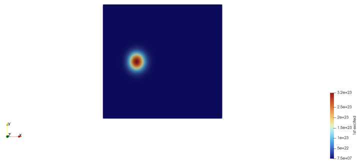
(c) Fourth excited level.

Figure 3.5: Separate line plots for the two quasi-degenerate pairs and for the fourth excited state.

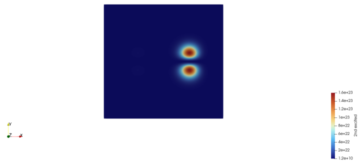
Figure 3.5 separates the same spectrum into physically corresponding groups. The first pair represents the lowest orbital localized in the two dots, while the second pair represents the first excited orbital in the two dots. The fourth excited state is different: its profile contains weight in both wells and in the central barrier. Since its energy is close to the barrier saddle, this state is the first one showing a clear molecular-like hybridization between the dots.



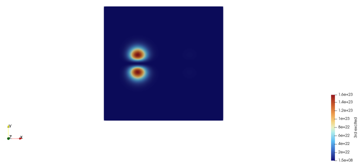
(a) Ground state.



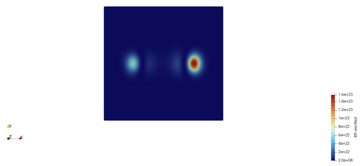
(b) 1st excited state.



(c) 2nd excited state.



(d) 3rd excited state.



(e) 4th excited state.

Figure 3.6: Slices of the probability densities $|\psi_i|^2$ in the quantum region.

The density slices in Fig. 3.6 confirm the assignment of the line profiles. The two lowest states are single-lobe states localized in opposite wells. The second and third excited states show the expected two-lobe structure of the first excited orbital, again localized mainly on one dot at a time. The fourth excited state is the only one among the computed levels with an appreciable density in the region separating the two wells.

The device therefore forms the expected double quantum dot: two confined potential minima, two nearly degenerate orbital pairs, and a finite inter-dot barrier. The lowest states correspond to a weakly coupled regime, because they are mostly localized on individual dots. The fourth state instead has a bonding-like hybridized character, evidenced by the probability density in the barrier region and by its energy lying close to the barrier saddle.